University Physics: Mechanics FINAL

Show equations, substitutions, intermediate reasoning, and a final labeled answer.

1. [8 pts] A cart starts at v■=8 m/s and accelerates at 2 m/s^2 for 5 s. Find final velocity and displacement; include units.

2. [8 pts] A 2 kg block is pulled horizontally by 13 N on a surface with μ_k=0.2. Draw the free-body diagram and compute acceleration (g=9.8 m/s^2).

3. [7 pts] A 3 kg object drops from height 6 m with negligible drag. Use energy conservation to find its speed just before impact.

4. [5 pts] State momentum conservation for a one-dimensional collision and identify the condition under which it applies.

5. [6 pts] A force-versus-time graph has a rectangular pulse of height 6 N lasting 0.5 s. Compute impulse and describe its effect on momentum.

6. [8 pts] A cart starts at v■=5 m/s and accelerates at 2 m/s^2 for 5 s. Find final velocity and displacement; include units.

7. [8 pts] A 7 kg block is pulled horizontally by 23 N on a surface with $\mu_k=0.2$. Draw the free-body diagram and compute acceleration ($g=9.8\ m/s^2$).

8. [7 pts] A 2 kg object drops from height 5 m with negligible drag. Use energy conservation to find its speed just before impact.

9. [5 pts] State momentum conservation for a one-dimensional collision and identify the condition under which it applies.

10. [6 pts] A force-versus-time graph has a rectangular pulse of height 6 N lasting 0.5 s. Compute impulse and describe its effect on momentum.